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1. Introduction & Definitions

GPWS gives visual and audible warnings when aircraft proximity to terrain poses a safety threat. Operates between **50 ft** and **2450 ft** actual height above surface. Automatically selects correct mode.

Term	Definition
Alert	A caution generated by the EGPWS equipment
Warning	A command generated by the EGPWS equipment
Genuine	Warning in accordance with technical specification
Nuisance	Alert in accordance with specification, but aircraft is flying an accepted safe procedure
False	System fault causes a warning not in accordance with specification

2. GPWS Modes — Summary Table

Mode	Condition	Alert	Warning
1	Excessive barometric descent rate	"Sink Rate"	"Whoop Whoop Pull Up"
2	Excessive terrain closure rate	"Terrain Terrain"	"Whoop Whoop Pull Up"
3	Altitude loss after T/O or G/A	"Don't Sink"	—
4A	Unsafe terrain clearance – gear not down	"Too Low Gear" / "Too Low Terrain"	—
4B	Unsafe terrain clearance – flaps not landing	"Too Low Flaps" / "Too Low Terrain"	—
5	Descent below ILS glide slope	"Glide Slope"	—
6A	Below selected minimum radio altitude	"Minimums Minimums"	—
6B	Altitude call-outs & bank angle	"Bank Angle"	—
7	Windshear	—	"Windshear"

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3. Mode 1 — Excessive Barometric Descent Rate

> Two boundaries. Independent of aircraft configuration. Operates 50–2450 ft RA.

- **First boundary:** Alert — "SINK RATE, SINK RATE" (repeated every **1.5 seconds**)
- **Second boundary:** Warning — "WHOOP WHOOP PULL UP" until rate of descent corrected
- **Visual:** PULL UP light

4. Mode 2 — Excessive Terrain Closure Rate

> Monitors Mach number, radio altitude rate of change, barometric altitude, and aircraft configuration.

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- **First boundary:** Alert — "TERRAIN, TERRAIN" + repeated warning "WHOOP WHOOP PULL UP"
- In landing configuration: only repeating "TERRAIN" alert (no PULL UP)
- Terrain message repeated every **1.5 seconds**
- Mach 0.35–0.45 with gear up: highest RA at which Mode 2 warning occurs increases to **2450 ft**
- Higher envelope inhibited with FLAP OVRD switch

5. Mode 3 — Altitude Loss after Take-off or Go-around

> Alerts if a descent is made during initial climb or go-around.

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- **Alert:** "DON'T SINK" (repeated every **1.5 seconds**)
- **Visual:** PULL UP light
- Effective between **50–700 ft RA**
- Activates when accumulated barometric loss ≈ **10%** of existing barometric altitude
- Does NOT arm during descent until below **200 ft RA**
- Arms when aircraft descends below 200 ft in landing configuration

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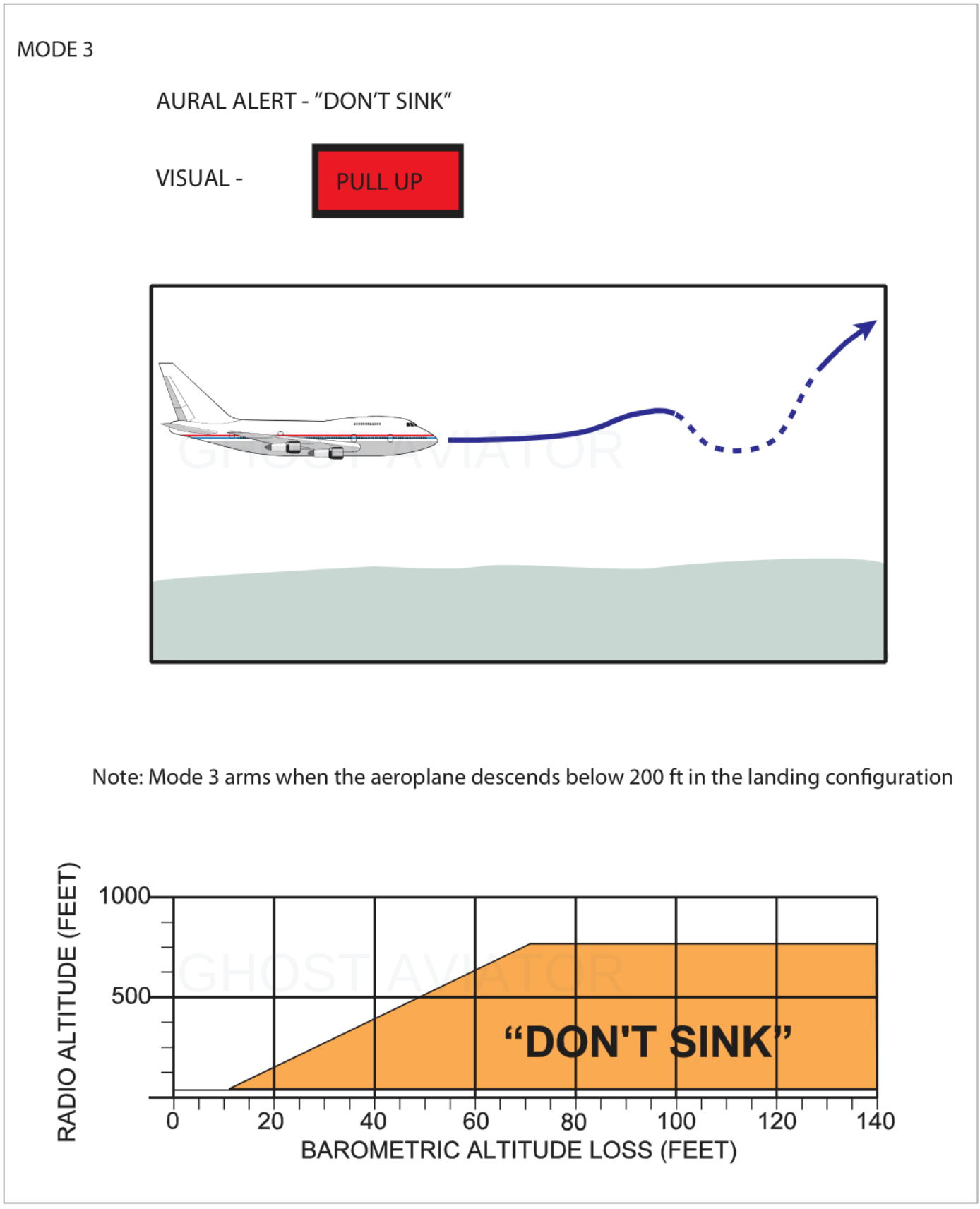


Fig 34.7 — GPWS Mode 3 altitude loss

6. Mode 4A — Unsafe Terrain Clearance (Gear Not Down)

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Armed after take-off upon climbing through **700 ft RA** with gear retracted.

- Below Mach 0.35: alert "TOO LOW GEAR"
- Above Mach 0.35: alert "TOO LOW TERRAIN" (upper boundary increases to **1000 ft RA**)
- Repeated every **1.5 seconds**
- **Visual:** PULL UP

7. Mode 4B — Unsafe Terrain Clearance (Flaps Not Landing)

Activates when gear is down and flaps are NOT in landing position.

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- Below Mach 0.28: alert "TOO LOW FLAPS"
- Above Mach 0.28: alert "TOO LOW TERRAIN" (upper boundary increases to **1000 ft RA**)
- "TOO LOW GEAR" takes priority over "TOO LOW FLAPS"
- Inhibited with FLAP OVRD switch

Mode 4 Speed Summary:

- Mode 4A: below Mach **0.35** = "TOO LOW GEAR"; above 0.35 = "TOO LOW TERRAIN"
- Mode 4B: below Mach **0.28** = "TOO LOW FLAPS"; above 0.28 = "TOO LOW TERRAIN"
- Both modes: arming altitude = **700 ft RA**; enhanced upper boundary = **1000 ft RA**

8. Mode 5 — Below Glide Slope Deviation Alert

Alerts crew of descent more than **1.3 dots** below ILS glide slope.

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- **Soft area:** Reduced volume "GLIDE SLOPE" (alert)
- **Hard area** (~2 dots below G/S): Full volume "GLIDE SLOPE GLIDE SLOPE"
- Voice repetition rate increases as deviation increases and RA decreases
- Armed when: valid G/S signal received + gear down + RA ≤ **1000 ft**

Mode 5 can be cancelled/inhibited by pressing either pilot's BELOW G/S light while below 1000 ft RA. Re-arms when climbing above 1000 ft RA.

Modes 1–4 aural alerts take priority over Mode 5 aural alerts. However, both PULL UP and BELOW G/S lights can illuminate simultaneously.

9. Mode 6A — Below Selected Minimum Radio Altitude

- Alert if descent below the minimum decision altitude cursor in the radio altimeter
- Operates between **50–1000 ft RA**
- **Aural only:** "MINIMUMS, MINIMUMS" — sounded once
- Re-arms when RA becomes greater than cursor-selected altitude

10. Mode 6B — Altitude Call-outs & Bank Angle Alert

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- Call-outs at selected altitudes (e.g., 200 ft and 100 ft to decision height) — customer option
- "BANK ANGLE" alert for excessive roll angles (airline-defined limits, not structural limits)
- Bank angle limit reduces with proximity to ground (reduced wingtip clearance to prevent wing tip/engine damage during T/O and landing)

11. Mode 7 — Windshear Alerting

Mode 7 warnings take priority over ALL other modes.

Multiple parameters monitored: ground speed, airspeed, barometric height, rate of descent, radio altitude. Warning generated when initial conditions of entering windshear are detected. Benefit: allows earlier initiation of the windshear go-around procedure, increasing probability of avoiding accident.

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GPWS/Mode 7 does NOT scan ahead — it detects windshear as the aircraft enters it. This is the key limitation compared to EGPWS forward-looking terrain awareness.

12. Action on Receiving GPWS Warnings

Immediate response to ALL GPWS warnings:

1. Level the wings
2. Initiate a maximum gradient climb
3. Maintain climb until minimum safe altitude for that route segment

Establishing cause takes second place.

Discontinuing the climb is permitted ONLY when:

- Cause positively identified and warning has ceased
- Aircraft operated by day in clear visual conditions
- Immediately obvious the aircraft is not in danger

Report ALL alerts/warnings to the operator — even genuine/nuisance. Repeated nuisance warnings may reduce crew confidence in the system, which reduces safety margins.

13. Enhanced GPWS (EGPWS) / TAWS

EGPWS adds a Terrain Awareness & Warning System (TAWS) — a forward-looking capability using a worldwide terrain database + INS/GPS position data.

Classic GPWS limitation: No "look ahead" — Mode 2 warning on high terrain requires the aircraft to already be over rising ground. A sheer cliff would give no Mode 2 warning until the aircraft is over it.

EGPWS Terrain Threat Display

- Shows terrain ahead of flight path in colour-coded display (green → yellow → red)
- Alert timing: **40–60 seconds** before predicted impact for caution (yellow); **20–30 seconds** for warning (red + solid)
- Messages: "Caution Terrain" / "Terrain Ahead" / "Obstacle Ahead" (if obstacle data) → "Whoop Whoop Pull Up"
- Threatening terrain: yellow (caution) → solid red (warning)

Misuse warning: Some pilots use the terrain display to "thread" through high terrain. This is a gross misuse and strongly advised against — display accuracy depends on navigation system accuracy.

The terrain display can be superimposed on the weather radar display, the ND, or on a dedicated PPI.

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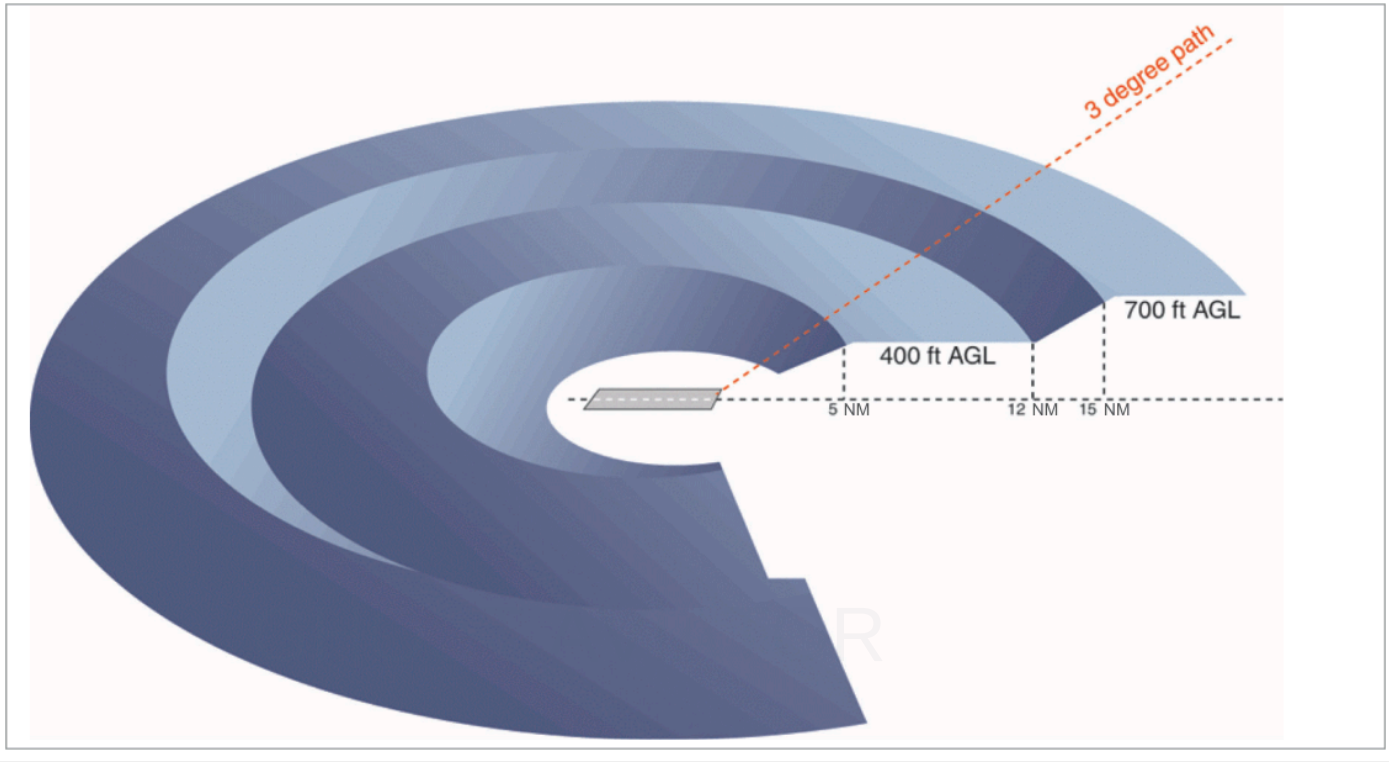


Fig 34.15 — Terrain clearance floor concept

14. Terrain Clearance Floor, BITE, Inhibition

Terrain Clearance Floor

Alerts crew to possible premature descent for non-precision approaches regardless of configuration. Uses aircraft position relative to the runway in the EGPWS database (separate from FMS, not subject to regular updates). Future integration with RAAS (Runway Awareness and Advisory System) planned.

Integrity Testing (BITE)

- BITE allows full testing of all functions and visual/aural warnings before flight
- Pre-flight BITE is inhibited in flight
- System continuously monitored in flight; failures automatically indicated
- Short confidence check possible in flight but not a full BITE check
- Test initiated by pressing the test switch

Inhibition of EGPWS Modes

> EGPWS must NOT be de-activated (CB pulled) except for approved procedures. Instructions must state no person may de-activate except per Operations Manual.

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Permitted inhibition cases:

- Mode 5 (G/S) — when G/S signal present but aircraft deliberately not using it (e.g., localizer-only approach, different runway)
- When gear or flap position inputs are known to be non-standard

Quick Revision Summary — Chapter 34:

- GPWS operates 50–2450 ft AGL. Mode 7 = highest priority
- Mode 1: Excessive barometric descent rate → "Sink Rate" / "Whoop Pull Up"
- Mode 2: Excessive terrain closure rate → "Terrain Terrain" / "Whoop Pull Up"; 2450 ft above Mach 0.45 gear up
- Mode 3: Altitude loss T/O or G/A → "Don't Sink"; 50–700 ft RA; 10% accumulated loss; arms below 200 ft
- Mode 4A: Gear not down; arms at 700 ft; <M0.35 "Too Low Gear"; >M0.35 "Too Low Terrain" (1000 ft)
- Mode 4B: Flaps not landing, gear down; <M0.28 "Too Low Flaps"; >M0.28 "Too Low Terrain" (1000 ft)
- Mode 5: >1.3 dots below G/S; soft/hard alert; armed G/S valid + gear down + ≤1000 ft RA; Modes 1–4 take priority
- Mode 6A: "Minimums Minimums" once; 50–1000 ft RA
- Mode 7: Windshear; takes priority over all other modes
- Action: Level wings + maximum gradient climb; cause second
- EGPWS/TAWS: forward-looking terrain database; 40–60 sec caution; 20–30 sec warning

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Practice Questions & Answers (Chapter 34)

Practice questions with verified answers.

Q1. The GPWS the alert/warning information is provided by a radio altimeter with:

a. A downward transmitting beam 60° and 30° (fore/aft and athwartship)

b. A downward transmitting beam 30° and 60° (fore/aft and athwartship)

c. A forward transmitting beam

d. A downwards transmitting radio beam

Correct Answer: **(d)** A downwards transmitting radio beam

Explanation: The basic GPWS RA uses a downward-transmitting beam to measure height above terrain. The specific beam dimensions (fore/aft 60°, athwartship 30°) describe the classic RA antenna pattern but the key answer is simply a downward radio beam. Reference answer key: d.

Q2. Mode 5 of GPWS is triggered when the aircraft is:

a. Below 500 ft RA with flaps not in landing position and speed below Mach 0.28

b. Below 500 ft RA with flaps not in landing position and speed below Mach 0.35

c. Below 200 ft barometric altitude with flap not in landing position and speed below Mach 0.28

d. Below 1000 ft RA and more than 1.3 dots below the ILS glide path

Correct Answer: **(d)** (reference key: b)

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Explanation: Mode 5 = glide slope deviation alert. Armed when: valid G/S signal + gear down + ≤1000 ft RA. Triggers when >1.3 dots below G/S. Note: the reference answer key shows **b** — which describes Mode 4B conditions (flaps/speed). The correct Mode 5 description is (d). This is a known numbering quirk in that question set — see Section 8 for Mode 5 details.

> ***Instructor's Note:*** *The reference answer key for this chapter gives Q3 = b. Mode 5 is: below 1000 ft RA, G/S valid, gear down, >1.3 dots below G/S. Know this clearly regardless of question phrasing.*

Q3. GPWS Mode 3 will give a "Don't Sink" alert during a go-around when altitude loss equals approximately:

a. 5% of existing barometric altitude

b. 10% of existing barometric altitude

c. 15% of existing barometric altitude

d. 200 ft absolute

Correct Answer: **(b)** 10% of existing barometric altitude

Explanation: Mode 3 activates when accumulated barometric loss equals approximately **10%** of the existing barometric altitude. See Section 5.

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Q4. Mode 4 (ground proximity with gear/flap not in landing configuration) — the arming altitude is:

a. 500 ft RA

b. 700 ft RA

c. 200 ft RA

d. 790 ft RA

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Correct Answer: **(b)** 700 ft RA

Explanation: Mode 4 (both 4A and 4B) is armed after take-off upon climbing through **700 ft RA**. See Sections 6 and 7.

Q5. GPWS Mode 7 (windshear warning) takes priority over:

a. Mode 5 only

b. Modes 5 and 6 only

c. All other modes

d. No priority — all modes equal

Correct Answer: **(c)** All other modes

Explanation: Mode 7 windshear warnings take priority over ALL other modes. Windshear is considered the highest-priority terrain/proximity threat because of its immediate effect on aircraft performance. See Section 11.

Q6. When the aircraft receives a GPWS warning, the FIRST action is to:

a. Identify the cause of the warning and report to ATC

b. Consult the QRH for the correct procedure

c. Level wings and initiate maximum gradient climb

d. Disengage the autopilot and pull up

Correct Answer: **(c)** Level wings and initiate maximum gradient climb

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Explanation: The immediate response to ALL GPWS warnings is: level the wings + maximum gradient climb. Establishing the cause takes second place. The response must be positive and immediate. See Section 12.

> ***Instructor's Note:*** *This is the critical operational rule. GPWS response drills: Level → Pull → Climb. Cause analysis comes after terrain clearance is assured.*

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